

Chapter 43 - Coprocessor Positive Cookbook

Chapter 32 explains the coprocessor registers and the failure path. This chapter shows the positive path: put a tiny worker image in guest RAM, start it, enqueue a request, wait for completion, inspect the shared mailbox, then stop the worker.

The worker in this chapter is a 6502 service. It does not use a hidden file. BASIC enters the 6502 bytes into memory and starts them with COPROC_CMD_START_MEM.

43.1 The Mailbox Shape

Each worker has a ring inside the shared mailbox at \$790000. The 6502 worker is CPU type 3, with cpuTypeIndex 1 and instance 0, so its ring index is $1 * 2 + 0 = 2$ at the uniform \$400 stride:

$$\$790000 + 2 * \$400 = \$790800$$

Inside the 6502 worker address space, that same ring appears at \$2800, because the worker maps the mailbox at CPU address \$2000.

Field	Bus address	6502 address
Ring head	\$790800	\$2800
Ring tail	\$790801	\$2801
Ring capacity	\$790802	\$2802
Request slot 0	\$790808	\$2808
Response slot 0	\$790A08	\$2A08
Response result code	\$790A10	\$2A10

The example assumes a fresh worker and sends one request, so slot 0 is enough.

43.2 The 6502 Service Bytes

This service waits until head and tail differ. It reads the request operation byte from request slot 0, adds 1, writes that value into the response descriptor result-code field, sets response status to COPROC_TICKET_OK, advances the tail to 1, and waits for more work.

AD 03 28 8D 04 28
AD 00 28 CD 01 28 F0 F8 AD 10 28 18 69 01 8D 10 2A
A9 00 8D 11 2A 8D 12 2A 8D 13 2A
8D 14 2A 8D 15 2A 8D 16 2A 8D 17 2A
A9 02 8D 0C 2A
A9 00 8D 0D 2A 8D 0E 2A 8D 0F 2A
A9 01 8D 01 28 4C 06 00

The same bytes as 6502 instructions:

```

$0000 LDA $2803      ; layout version
$0003 STA $2804      ; acknowledge version
$0006 LDA $2800      ; head
$0009 CMP $2801      ; tail
$000C BEQ $0006
$000E LDA $2810      ; request op, low byte
$0011 CLC
$0012 ADC #$01
$0014 STA $2A10      ; response result code low byte
$0017 LDA #$00
$0019 STA $2A11
$001C STA $2A12
$001F STA $2A13
$0022 STA $2A14      ; response length = 0
$0025 STA $2A15
$0028 STA $2A16
$002B STA $2A17
$002E LDA #$02
$0030 STA $2A0C      ; response status = OK
$0033 LDA #$00
$0035 STA $2A0D
$0038 STA $2A0E
$003B STA $2A0F
$003E LDA #$01
$0040 STA $2801      ; tail = 1
$0043 JMP $0006

```

43.3 Type The Positive Example

This BASIC listing enters the service bytes, starts the worker from guest RAM, enqueues operation 7, waits for completion, and inspects the response descriptor.

```

10 REM 6502 COPROCESSOR POSITIVE PATH
20 S=MEMALLOC(70,16)
30 FOR I=0 TO 69:READ B:POKE8 S+I,B:NEXT
40 POKE32 &H000F2344,3
50 POKE32 &H000F235C,S
60 POKE32 &H000F2360,70
70 POKE32 &H000F2340,6
80 PRINT "START ";PEEK32(&H000F2348),PEEK32(&H000F234C)
90 REQ=MEMALLOC(4,4):RESP=MEMALLOC(4,4)
100 POKE32 REQ,&H12345678
110 T=COCALL(3,7,REQ,4,RESP,4)
120 PRINT "TICKET ";T
130 COWAIT T,1000
140 PRINT "STATUS ";COSTATUS(T)
150 PRINT "RESULT ";PEEK32(&H00790A10)
160 COSTOP 3
170 DATA &HAD,&H03,&H28,&H8D,&H04,&H28,&HAD,&H00
180 DATA &H28,&HCD,&H01,&H28,&HF0,&HF8,&HAD,&H10
190 DATA &H28,&H18,&H69,&H01,&H8D,&H10,&H2A,&HA9
200 DATA &H00,&H8D,&H11,&H2A,&H8D,&H12,&H2A,&H8D
210 DATA &H13,&H2A,&H8D,&H14,&H2A,&H8D,&H15,&H2A
220 DATA &H8D,&H16,&H2A,&H8D,&H17,&H2A,&HA9,&H02
230 DATA &H8D,&H0C,&H2A,&HA9,&H00,&H8D,&H0D,&H2A
240 DATA &H8D,&H0E,&H2A,&H8D,&H0F,&H2A,&HA9,&H01
250 DATA &H8D,&H01,&H28,&H4C,&H06,&H00

```

Expected result: START is followed by status 0 and command error 0, TICKET is non-zero, STATUS is 2, and RESULT is 8. The exact ticket number is not fixed. It only has to be non-zero. The result is 8 because the request operation was 7 and the worker adds 1.

43.4 What The Lines Do

Lines 20 to 30 allocate public low memory and enter the 6502 image. Lines 40 to 70 start a CPU type 3 worker from that memory by writing the raw coprocessor MMIO command 6.

The service's first two instructions copy the ring layout version from \$2803 to \$2804. This acknowledgement is required before the start command succeeds and before the manager routes requests to the worker.

The stores from \$0014 through \$003B finish the response result, length, and status fields. The store at \$0040 advances `tail` only after those fields are complete. That final byte is the publication step: once the caller sees the new tail, it may consume the response. Moving the tail store earlier could expose a response whose later fields still belong to the preceding operation. The ordering of these stores is what makes the complete descriptor ready.

Line 110 enqueues a normal BASIC COCALL. BASIC checks that REQ and RESP are public buffers created by MEMALLOC.

Line 130 waits for the worker. Line 140 prints the ticket status: 2 means COPROC_TICKET_OK. Line 150 reads the response descriptor inside the mailbox. This example uses the descriptor result code rather than a response byte buffer, because that keeps the 6502 service short and makes the mailbox visible to the reader.

43.5 Side Effects and Limits

- A running worker remains online until COSTOP or until the worker exits.
- COCALL returns ticket 0 if enqueue fails. Read COPROC_CMD_ERROR at \$F234C for the command error.

- COWAIT uses the timeout in milliseconds. If it times out, `COSTATUS(ticket)` reports 4.
- This first service handles the first request slot only. A full service must use the current tail value to choose a request and response slot.
- The 6502 worker sees its own 64 KB service memory plus the mailbox window. It does not automatically see every BASIC buffer by 16-bit address.

Chapter 44 shows how to debug this kind of worker with IE Mon and IE Script.